Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/22/2015
PI Name:	Cheng, Zhengdong Ph.D.		
Project Title:	Liquid Crystals of Nanoplates		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Liquid Crystal Facility		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AQ60G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The central objectives of the proposed research are to investigate the dynamics of phase transitions, including the formation of liquid crystals as well as the equilibrium gel or empty liquid state, of highly anisotropic colloidal disks. We offer unprecedented control over the size (20 nm to 1 micron), thickness (a few to hundreds of layers), and their polydispersities (14% to 30%) of these disks via the controlled growth of layered inorganic crystals. The nanoplates obtained via exfoliation of the crystals have identical thickness of the order of 1 nm and controlled size and size distribution. We plan to use the Advanced Colloids Experiment (ACE) microscope in microgravity environment to observe the kinetics of liquid crystal (the nematics, chiral nematics, smectics, and the twist grain boundary phase) nucleation and growth, and kinetics of gelation phase separation. The comparison studies on the growth of the chiral crystals in 1g and mciro-g will reveal the role of gravity in alignment of the liquid crystal domains. The proposed		

	research extends the existing ACE investigations to understand the full phase transitions of discotic colloids by the capability to fabricate disks/nanoplates with controlled aspect ratios. It will offer novel colloidal self-assembly structures including the chiral nematics, the smectics, the twist grain boundary (TGB) phase, and the equilibrium gel or empty liquid state. The understanding of the structure and properties of colloidal disks are important to soil science and space exploration (e.g., Mars), to the fundamental investigation of discotic liquid crystals, as well as various industries, including oil recovery, the delivery of pharmaceuticals, nutrition, and cosmetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The understanding of the structure and properties of colloidal disks are important to soil science and space exploration (e.g., Mars), to the fundamental investigation of discotic liquid crystals, as well as various industries, including oil recovery, the delivery of pharmaceuticals, nutrition, and cosmetics.
Task Progress:	1. Kinetics of nucleation, growth and sedimentation: The kinetics of formation of nematic phase from a well-mixed liquid suspension proceeds via the nucleation and growth of nematic tactoids and their sedimentation. Microscopic observation reveals that the tactoids change shape as they grow in size. Initially homogeneous director field is observed in small tactoids, but as they grow bigger radial director field and other distorted patterns of director are assumed by tactoids. Due to gravity, the growth is not complete and the tactoids sediment. In microgravity environment, the sedimentation will be negligible and growth of tactoids and corresponding director field inside tactoids will be investigated. 2. Growth of single domain nematics in gravity: After initial isotropic-nematic phase separation, the nematic phase is multi-domain as it shows multi-grain texture when observed between crossed polarizers. After a long time, as long as a year, the nematic phase develops into single domain phase. The hypothesis of this single domain phase being a biaxial nematic is to be tested. 3. Shear induced birefringence in liquid (isotropic) suspensions: The equilibrium isotropic phase does not show any birefringence at equilibrium and hence appears dark when observed between crossed polarizers. But when given a shear, the isotropic suspension shows birefringence because of the temporary ordering of nanoplates due to flow. As the suspension gets more concentrated, the decay of flow induced birefringence becomes slower. We studied this phenomena with different size of nanoplates with identical thickness. Larger size also favors slower decay. 4. Investigation of Jammed Gel phase: The jammed gel phase due to the face-to-rim attraction are also enabled in the phase diagram, showing the fluid-to solid transition via weak attraction from face to rim of the a-ZrP nanoplates. Due to the non-homogeneity of charge distribution at face and rim, ZrP nanoplates has the specific face-to-rim interaction. Due to the difference in isoelectric point on the face and rim of a -ZrP crystal, the interaction of face and rim can be attractive in nature at low pH or repulsive at high pH. Owing to its ionic exchange behavior, a cation exchange with proton on a -ZrP crystal surface decreases the pH. With increasing the NaCl/ZrP molar ratio in the suspension, there are more protons that get dragged to the surrounding medium to decrease the pH, where gelation occurs, so that it forms gel at low pH while liquid crystal phase remains at high pH. However, it is also possible to have a gel phase pH near neutral by means of depletion attraction between nanoplates induced by polymer chain. When ion exchange polymer sodium poly acrylic acid (PAANa) is added to a-ZrP suspension in water, proton exchange with sodium occurs locally to obtain T-shaped structure. Here, PAA polymeric chain serves as a buffer solution to keep suspension stay neutral even at salt to nanoplate ratio as high as 5. 5. Investigation of Equation of State of Colloidal Nanoplates: In order to study the equation of state of colloidal nanoplates, i.e., osmotic pressure vs. volume fraction relationship, we made use of gravity to induce concentration gradient in a vertical tube. X-ray densitometry technique is used for obtaining density of nanoplates at different heights and then osmotic pressure is calculated by making use of the fact that osmotic pressure gradient balances the effective weight of nanoplates. Our approach has not led us to find the equation of state since the density gradient in nematic phase does not follow the theoretical prediction. This could be because the system of nematic colloids may not have reached equilibrium at the time of experiment. Moreover size fractionation in the isotropic phase is much of possibility due to polydisperse nature of the suspension and is seen from the steep increase in osmotic pressure in isotropic phase as the I-N interface is approached. In future, in house measurement system is to be setup to continuously monitor the equation of state.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
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